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Microelectronic devices and related methods of fabricating microelectronic devices

Abstract

A microelectronic device may have side surfaces each including a first portion and a second portion. The first portion may have a highly irregular surface topography extending from an adjacent surface of the microelectronic device. The second portion may have a less uneven surface extending from the first portion to an opposing surface of the microelectronic device. Methods of forming the microelectronic device may include creating dislocations in the wafer in a street between the one or more microelectronic devices by implanting ions and cleaving the wafer responsive to failure of stress concentrations near the dislocations through application of heat, tensile forces or a combination thereof. Related packages and methods are also disclosed.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 17/241,386, filed Apr. 27, 2021, now U.S. Pat. No. 11,784,050, issued Oct. 10, 2023, the disclosure of which is hereby incorporated herein in its entirety by this reference.

TECHNICAL FIELD

(1) Embodiments of the disclosure relate to a method of fabricating microelectronic devices. Specifically, embodiments relate to methods of separating wafers comprising arrays of microelectronic device locations into individual microelectronic devices, and to related microelectronic devices, tools, and apparatus.

BACKGROUND

(2) As performance of electronic devices and systems increases, there is an associated demand for improved performance of the microelectronic components of such systems, while maintaining or even shrinking the form factor (i.e., length, width and height) of a microelectronic device or assembly. Such demands are often, but not exclusively, associated with mobile devices and high-performance devices. To maintain or reduce the footprint and height of an assembly of components in the form of microelectronic devices (e.g., semiconductor dice), three-dimensional (3D) assemblies of stacked components equipped with so-called through silicon vias (TSVs) for vertical electrical (i.e., signal, power, ground/bias) communication between components of the stack have become more common, in combination with the reduction in component thickness, as well as employment of preformed dielectric films in the bond lines (i.e., spaces between stacked components) to reduce bond line thickness while increasing bond line uniformity. Such dielectric films include, for example, so-called non-conductive films (NCFs), and wafer level underfills (WLUFs), such terms often being used interchangeably. While effective in reducing height of 3D microelectronic device assemblies, the reduction in thickness of microelectronic devices, for example semiconductor dice, to about 50 μm or less (e.g., 30 μm , 20 μm) increases device fragility and susceptibility to cracking under stress, particularly compressive (i.e., impact) stress and

bending stress. Decreasing bond line thickness may also exacerbate susceptibility to damage to such extremely thin microelectronic devices, as the thin dielectric material (e.g., NCF) in the bond lines may no longer provide any cushioning effect or ability to accommodate particulate contaminants in the bond lines when, for example, a device is stacked on another device to form a 3D assembly. Non-limiting examples of microelectronic device assemblies including stacked microelectronic devices which may suffer from stress-induced cracking include assemblies of semiconductor memory dice, alone or in combination with other die functionality (e.g., logic) include so-called high bandwidth memory (HBMx), hybrid memory cubes (HMCs), and chip to wafer (C2 W) assemblies.

(3) Furthermore, as the demand for microelectronic devices increases, the demand for lower cost microelectronic devices also increases, incentivizing the continued increase in circuit density and devices per wafer. The cost of producing microelectronic devices may be reduced by increasing efficiency of the processes, increasing the yield of microelectronic devices per wafer for each respective process and reducing losses, such as due to circuit failures, physical die cracking, microcracking, and fractures, etc. Decreasing the cost of the microelectronic devices may, in turn, decrease the cost of the associated electronic assemblies and systems incorporating such microelectronic devices. In some cases decreasing the cost of the microelectronic devices may also enable increases in performance of the associated electronic devices without being cost prohibitive.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) While the specification concludes with claims particularly pointing out and distinctly claiming embodiments of the present disclosure, the advantages of embodiments of the disclosure may be more readily ascertained from the following description of embodiments of the disclosure when read in conjunction with the accompanying drawings in which:

(2) FIG. 1 illustrates a schematic view of a microelectronic device assembly comprising stacked semiconductor dice;

(3) FIG. 2 illustrates a perspective view of a wafer with an array of microelectronic device locations formed thereon;

(4) FIGS. 3-8 illustrate cross-sectional views of process acts of forming microelectronic devices in accordance with an embodiment of the present disclosure;

(5) FIG. 9 illustrates a plan view of a side surface of a microelectronic device in accordance with an embodiment of the present disclosure;

(6) FIG. 10 illustrates a profile view of a side surface of a microelectronic device in accordance with an embodiment of the present disclosure;

(7) FIG. 11 illustrates a top view of a microelectronic device in accordance with an embodiment of the present disclosure; and

(8) FIG. 12 illustrates a schematic side view of a microelectronic device package comprising a stack of microelectronic devices in accordance with an embodiment of the present disclosure.

DETAILED DESCRIPTION

(9) The illustrations presented herein are not meant to be actual views of any particular microelectronic device, assembly, or component thereof, but are merely idealized representations employed to describe illustrative embodiments. The drawings are not necessarily to scale.

(10) As used herein, the term “substantially” in reference to a given parameter means and includes to a degree that one skilled in the art would understand that the given parameter, property, or condition is met with a small degree of variance, such as within acceptable manufacturing tolerances. For example, a parameter that is substantially met may be at least about 90% met, at least about 95% met, at least about 99% met, or even at least about 100% met.

(11) As used herein, relational terms, such as “first,” “second,” “top,” “bottom,” etc., are generally used for clarity and convenience in understanding the disclosure and accompanying drawings and do not connote or depend on any specific preference, orientation, or order, except where the context clearly indicates otherwise.

(12) As used herein, the term “and/or” means and includes any and all combinations of one or more of the associated listed items.

(13) As used herein, the terms “vertical,” “horizontal,” and “lateral” refer to the orientations as depicted in the figures.

(14) As the demand for microelectronic components increases, the demand for lower cost microelectronic components also increases. The cost of producing microelectronic devices may be reduced by increasing the yield of microelectronic devices for each act performed in the fabrication process. Yield may be increased by reducing losses, such as failed devices, broken devices, etc. Another way that the yield may be increased is by reducing the area of a semiconductor wafer that is required between adjacent microelectronic device locations. For example, reducing a width of the so-called “street” or “scribe” area between adjacent, individual microelectronic device locations where the semiconductor wafer is singulated or diced to separate the semiconductor wafer into separate microelectronic devices may increase the area of the semiconductor wafer that is available for forming the microelectronic devices. Thus, decreasing the street width between the individual microelectronic devices may enable a larger number of microelectronic devices to be formed from each semiconductor wafer.

(15) FIG. 1 illustrates a microelectronic device **100**. The microelectronic device assembly **100** may include multiple semiconductor dice **102** arranged in a stack. A dielectric film **104**, such as a non-conductive film (NCF) or a wafer level underfill (WLUF) may be positioned in a so-called bond line between each of the semiconductor dice **102**. The microelectronic device **100** may include through silicon vias (TSVs) **106** aligned with contacts in the form of conductive pillars **108p** optionally capped with solder material **108s** and bonded to terminal pads **108t** of adjacent semiconductor dice **102** to provide electrical contacts between the semiconductor dice **102** and/or through the stack of dice. For example, the TSVs **106** and aligned contacts may provide power, ground/bias, and signal connections.

(16) The height of the microelectronic device **100** may be reduced by reducing a thickness of the semiconductor dice **102** and/or the dielectric film **104**. Reducing the thickness of the semiconductor die **102** may cause a semiconductor die **102** to be more fragile and susceptible to damage in the form of microcracking, cracking and edge chipping during the picking and stacking processes as described in further detail below. Reducing the thickness of the dielectric film **104** may reduce the ability of the dielectric film **104** to provide any cushioning effect between adjacent semiconductor dice **102** during stacking, as well as the ability to accommodate particulate contaminants generated during separation of the semiconductor dice **102** in bond lines thinner than the particle size without damage to the semiconductor dice **102**. For example, contaminant particles between the semiconductor dice **102** may cause one or more of the semiconductor die **102** to crack or microcrack due to stress concentrations caused by the presence of contaminant particles larger than a thickness of dielectric film **104** when a semiconductor die **102** is picked from a carrier, transferred to a bond tip, or stacked on another semiconductor die **102** or a substrate. Further, in some instances, contaminant particles between the semiconductor die **102** may substantially prevent one or more of the conductive pillars **108p** from making electrical contact with an aligned terminal pad **108t**, or compromise integrity of such contact.

(17) Reducing and/or eliminating the introduction of contaminant particles to surfaces of the semiconductor dice during the singulation process may increase a yield of microelectronic devices by substantially reducing the losses from damage caused by such particles. Increasing the yield of microelectronic devices may decrease the costs associated with producing the microelectronic devices. These reduced costs may similarly reduce costs of associated electronic products

incorporating the microelectronic devices, such as mobile phones, computers, laptops, etc.

(18) Some embodiments of the present disclosure may include a method of fabricating a microelectronic device, including forming an array of microelectronic device locations on an active surface of a wafer. The method may further include securing the wafer to a carrier wafer. The method may also include thinning the wafer to about 30 microns (μm) or less.

(19) The method may further include implanting ions to initiate dislocations in semiconductor material of the wafer along streets between the microelectronic devices. The method may also include heating the wafer to form cracks along the streets from stress concentrations proximate the dislocations.

(20) During the fabrication process, semiconductor dice may be formed in an array on a wafer. The locations of individual dice **204** may be formed on an active surface **208** of a wafer **202**. The wafer **202** may be formed from a semiconductor material, such as silicon configured to provide a substrate for fabrication of integrated circuitry as well as structural support to the dice **204**. The dice **204** may be formed by building up layers of insulating and conductive materials on the active surface **208** of the wafer **202** through processes, such as plating, sputtering, etc. The process of forming the dice **204** may also include material removal processes, such as wet etching, dry etching, photolithography, etc. The material removal and/or build up processes may utilize masks to control where material is removed and/or built up. The building up and removal processes may form features of the microelectronic devices, such as vias, through silicon vias (TSVs), wiring paths, under bump metallization (UBM), etc.

(21) Each die **204** may be separated from adjacent dice **204** by streets **206**. A width of the streets **206** may define distances between the dice **204**. When using conventional singulation processes, reductions in width of the streets **206** may be constrained by the width required by the tool (e.g., wafer saw, laser beam, etc.) used to singulate (e.g., dice, cut, separate) the wafer **202** into individual dice **204**. Conventional blade dicing operations may require the width of the streets **206** to be between about 20 micrometers (μm) and about 80 μm , or even more, depending on blade width. Thus, a significant portion of the surface of the wafer **202** must be dedicated to the area of the streets **206** between the dice **204**. Reducing the width of the streets **206** may enable a larger number of dice **204** to be formed on each wafer **202**, which may increase the yield for each wafer **202**.

(22) After the dice **204** are formed on the active surface **208** of the wafer **202**, the wafer **202** may be coupled to a carrier wafer **304** through an adhesive **306**, as illustrated in FIG. 3. The adhesive **306** may couple the active surface **208** of the wafer **202** to the carrier wafer **304**, such that a rear surface **302** of the wafer **202** remains exposed. The adhesive **306**, may be an adhesive material configured to secure the wafer **202** at elevated temperatures above ambient (e.g., about 25° C.) such as temperatures between about 150° C. and about 250° C., such as between about 170° C. and about 220° C., or between about 180° C. and about 200° C. Some examples of adhesives that are formulated to secure a wafer at elevated temperatures may include BREWERBOND® materials sold by BREWER SCIENCE® of Missouri, the TA series of adhesives sold by SHIN-ETSU® of Taiwan, or the XP series of adhesives sold by DOW CHEMICAL® of Michigan. The carrier wafer **304** may be configured to support the wafer **202** through additional prior processing acts, such as wafer thinning. For example, very thin wafers **202** may require exceptionally rigid support to substantially prevent cracking, warping, and other potential damage to the wafer **202** during processing acts that may include temperature changes, mechanical material removal, chemical material removal, etc.

(23) Once the wafer **202** is adhered to the carrier wafer **304**, the wafer **202** may be thinned as illustrated in FIG. 4. The wafer **202** may be thinned from an initial thickness, for example between about 775 μm and about 600 μm through material removal processes, such as back grinding, polishing processes and wet etching. The material removal processes may remove material from the exposed rear surface **302** of the wafer **202**. In some embodiments, a latter portion of the material removal process (i.e., wet etching) may be used to remove sufficient semiconductor

material to expose features, such as TSVs through the rear surface **302** of the wafer **202** to form electrical connections between the rear surface **302** and integrated circuitry of the active surface **208** of the wafer **202**.

(24) The material removal process may thin the wafer **202** to a thickness of less than about 50 μm , such as less than about 30 μm , less than about 20 μm , or less than about 10 μm . Reducing the thickness of the wafer **202** may in turn reduce the thickness of the resulting microelectronic devices. For example, the resulting microelectronic devices may have a thickness of between about 30 μm and about 8 μm , such as between about 20 μm and about 7 to 10 μm . In the latter case, the microelectronic devices may include integrated circuitry to a depth of about 5 to 8 μm , and a thickness of supporting semiconductor material of about 2 μm . While the wafer **202** is being processed, one or more implantation (i.e., dopant) species may be implanted into the streets **206** between the dice **204** in an implant process, as illustrated in FIG. 5. FIG. 5 illustrates the implant process occurring on the rear surface **302** of the wafer **202** after the thinning process illustrated in FIG. 4; however, it is noted that the implantation process may be performed at earlier stages and/or on the active surface **208** of the wafer **202** as will be described in detail herein below.

(25) During implantation, ions of an implantation species, such as hydrogen, helium, arsenic, boron, phosphorus, etc., may be accelerated toward a surface of the wafer **202**. An implant tool **506** may include a beam generator **508** configured to accelerate the ions and form an energy beam **512** of ions to impinge on the surface of the wafer **202**. The beam generator **508** may receive ions of the respective species from an ion source. The ions may then be accelerated to a high energy through an electrostatic accelerator, such as a magnet or field of magnets. After being accelerated, the ions may have an energy of at least about 10 kiloelectronvolts (keV), such as at least about 100 keV, or at least about 250 keV. The implant tool **506** of the present disclosure may be configured to implant ions into the wafer **202** without the aid of an increased temperature, commonly referred to in the art as a temperature drive. Not increasing the temperature above ambient (e.g., about 25° C.) may allow the implant process to be performed after the dice **204** have been formed without compromising the thermal budget of the dice **204** and causing damage to the integrated circuitry and associated features of the dice **204**. Furthermore, not increasing the temperature may allow the implant process to be performed on the wafer **202** after the thinning process without risk of warping or otherwise damaging the wafer under excessive heat.

(26) The energy imparted to the ions during implantation may affect the depth into the wafer **202** that the ions will penetrate. For example, ions having higher energy may penetrate the wafer **202** to a greater depth than ions of the same species having a lower energy. The ion species may also effect the penetration depth of the ions. For example, ions of a small molecular species, such as hydrogen or helium, may have a greater penetration depth than ions of a larger molecular species, such as boron, phosphorus, or arsenic. In some embodiments, multiple different implant species may be used on the same wafer **202**, such that multiple different depths of penetration may be achieved in the same regions of the wafer **202**. Penetration of the ions of the implantation species into the semiconductor material of the wafer **202** may cause damage to the wafer **202**, such as point defects, dislocations, etc., at the penetration depth and/or along the path to the penetration depth. During the implant process the ions may penetrate to depths of at least about 2 μm , such as at least about 3 μm , at least about 4 μm . The smaller species of ions may penetrate to greater depths, such as at least about 4 μm and may cause smaller amounts of damage during the penetrations. On the other hand, while the larger species of ions may not penetrate the wafer to as great of a depth as the smaller species, the larger species of ions may generate larger amounts of damage at the lesser depth than the smaller species. In one implementation, hydrogen or helium ions may first be implanted to an ultimate implantation depth, after which boron, phosphorous or arsenic ions may be implanted above and to a lesser depth than the hydrogen or helium molecules.

(27) A mask **502** may be positioned between the implant tool **506** and the wafer **202**. The mask **502** may be configured to control and limit the portions of the wafer **202** on which the ions impinge.

For example, the mask **502** may include a pattern of openings **504**. The openings **504** may be arranged such that the ions only impinge on the portions of the surface of the wafer **202** that coincide with the streets **206** between the dice **204**.

(28) In some embodiments, the mask **502** may be positioned over the wafer **202**, such as through a mask aligner or stepper, etc. For example, the mask **502** may be aligned with the rear surface **302** of the wafer **202**, as illustrated in FIG. 5, such that the openings **504** in the mask **502** are substantially aligned with the streets **206** on the active surface **208** of the wafer **202**. In some embodiments, the mask **502** may be formed on the active surface **208** of the wafer **202** during the die fabrication process discussed with respect to FIG. 2. For example, the mask **502** may be positioned or formed on the active surface **208** of the wafer **202** and the ions may be implanted into the streets **206** between the die **204** by the implant tool **506** after the dice **204** are formed and before the active surface **208** of the wafer **202** is coupled to the carrier wafer **304**. In some embodiments, the mask **502** may be one of the masks used in a material adding or material removal process as discussed above with respect to FIG. 2. The implant tool **506** may then implant the ions into the active surface **208** of the wafer **202** during, directly before or after one of the material build up or material removal processes.

(29) In some embodiments, a material removal process, such as dry etching may be used to remove material in the streets **206** and form trenches or channels in the area of streets **206** before the implant process. Removing material in the street **206** before the implant process may enable the implanted ions to penetrate a greater distance into the wafer **202** from the active surface **208** before the material removal process. In some embodiments, the material removal process and the implant process may utilize the same mask **502**.

(30) In some embodiments, the mask **502** may be coupled to the implant tool **506**. For example, the mask **502** may be a reusable mask **502** configured for a specific die type (i.e., length and width) coupled to a face **510** of the implant tool **506**, or may be mounted internally within the tool chamber. The implant tool **506** may then be substantially aligned with the wafer **202**, such that the openings **504** in the mask **502** are substantially aligned with the streets **206**. The implant tool **506** may then implant ions into the active surface **208** and/or the rear surface **302** of the wafer **202** through the openings **504** in the mask **502**.

(31) The openings **504** in the mask **502** may have a width corresponding to a width of desired separation along streets between locations of dice **204** of less than about 10 μm , such as between about 1 μm and about 10 μm , or between about 1 μm and about 5 μm , or between about 1 μm and about 2 μm .

(32) FIG. 6 illustrates the wafer **202** after the thinning process of FIG. 4 and the implant process of FIG. 5. After the implant process the wafer **202** may include an implanted region **602**. The implanted region **602** may include molecules of the implant species as well as dislocations and point defects associated with the implant process. As a result of implanting the wafer **202** through the mask **502**, the implanted region **602** may substantially coincide with the openings **504** in the mask **502**. Therefore, the implanted region **602** may substantially coincide with the streets **206** between the dice **204**. The implanted region **602** may include a concentration of implant species of between about $1\text{e}12$ atoms/cm.^{sup.3} and about $10\text{e}16$ atoms/cm.^{sup.3}, such as between about $1\text{e}12$ atoms/cm.^{sup.3} and about $5\text{e}16$ atoms/cm.^{sup.3}.

(33) Larger concentrations of the implanted ion species may result in greater amounts of damage to the wafer **202** in the implanted regions **602**. As the implanted regions **602** may coincide with the streets **206** between the dice **204**, the damage to the wafer **202** may be substantially concentrated within the streets **206**. As noted above, openings **504** in the mask **502** may be sized, such that the implanted regions **602** may have a width **604** of less than about 10 μm , such as between about 1 μm and about 5 μm , or between about 1 μm and about 2 μm .

(34) The implanted regions **602** may have a greater amount of damage and residual implanted ions proximate the active surface **208**, or rear surface **302** of the wafer **202** into which the ions were

implanted by the implant tool **506**. The damage and/or residual implanted ions may gradually reduce as the depth into the wafer **202** increases.

(35) In some embodiments, as illustrated in FIG. 7, the wafer **202** may be heated while coupled to the carrier wafer **304**. Heating the wafer **202** may cause the wafer **202** to fracture in the implanted regions **602**. For example, the damage caused by the implanted ions in the implanted regions **602** may create stress concentrations within the wafer **202**. As the wafer **202** is heated the stress concentrations in the implanted regions **602** may cause cracks **702** to propagate through the thickness of the wafer **202** in an area substantially aligned with the implanted regions **602** and cleave the semiconductor material along the area of streets **206**. The cracks **702** may substantially separate (e.g., singulate) the wafer **202** into individual dice **204**. Heating the wafer **202** above ambient (e.g., about 25° C.) may cause the temperature of the wafer **202** to rise between about 150° C. and about 250° C., such as between about 170° C. and about 220° C., or between about 180° C. and about 200° C. At higher concentrations of ions as well as damage in the implanted regions **602**, relatively lower temperature changes may generate the cracks **702** and singulate the wafer **202**. Similarly, deeper penetration may enable the cracks **702** to be generated at lower temperatures. Thus, implanting ions of a smaller species, such as hydrogen or helium, may enable the cracks **702** to propagate at lower temperatures, which may reduce the time and energy required to singulate the wafer **202** and may reduce the risks of temperature damage to the wafer **202** and associated dice **204**. It may also be desirable to implant smaller species of ions, for example, hydrogen or helium, in implementation of this embodiment in light of an obtainable deeper penetration depth at a reasonable energy level in comparison to larger species. In addition, singulating at lower temperatures significantly reduces the impact on the thermal budget of semiconductor dice, lessening any potential for degradation of the integrated circuitry.

(36) The wafer **202** may be heated using heating tools, such as hot plates, hot chucks, lasers, resistance heaters, etc. For example, the carrier wafer **304** and the wafer **202** may be placed on a hot plate or hot chuck. In some cases, a hot plate or hot chuck may contact the rear surface **302** of the wafer **202**. In other embodiments, the beam of a laser may impinge on a surface **208**, **302** of the wafer **202** or a resistance heater may be placed over wafer **202** in close proximity to cause the temperature of the wafer **202** to increase. Heating wafer **202** from above may result in a smaller temperature gradient between the heat source and the implant area.

(37) During the heating process the carrier wafer **304** and adhesive **306** may, in combination support the wafer **202** against any displacement. For example, the carrier wafer **304** may enable the wafer **202** to absorb the heat after thinning without any substantial heat damage, such as warping, or shifting of position of the semiconductor dice **204** singulated from the wafer. Further, singulation on the carrier wafer **304** and picking the semiconductor dice **204** from the carrier wafer after release of the adhesive **306** eliminates potential damage to wafer **202** and semiconductor dice **204** resulting from transfer to an expandable carrier material (e.g., dicing tape, mount tape or film) for a subsequent pick and place operation of the semiconductor dice **204** with a pick tool, as well as potential damage from debond of semiconductor dice **204** from adhesive securing the dice to the carrier material. The adhesive **306** may be formulated to release when exposed to light, such as in the infrared (IR) range or the ultraviolet (UV) range. Light in the IR or UV range may penetrate through the carrier wafer **304** causing the adhesive **306** to release.

(38) In some embodiments, the semiconductor dice **204** may be picked from the surface of the carrier wafer **304** with a die stacking tool, such as a bond head, configured to lift a semiconductor die **204** from the rear surface **302**. The die stacking tool may be configured to clean the active surface **208** of the semiconductor die **204**, to clear any residual adhesive and/or particles from the active surface **208**. For example, the die stacking tool may include a nozzle for cleaning the active surface **208** of the semiconductor die **204**. In some embodiments, the semiconductor die **204** may be transferred to a cleaning tool, such as a cleaning platform configured to clean the residual adhesive from the active surface **208**. The die stacking tool may then be configured to couple the

semiconductor die **204** to other semiconductor dice forming a die stack, as discussed in further detail below.

(39) Some embodiments of the present disclosure may include a method of separating microelectronic devices (e.g., semiconductor dice) from a wafer. The method may include employing an ion implantation process to provoke dislocations in a semiconductor wafer in streets between the microelectronic devices. The method may further include transferring the semiconductor wafer to a laterally expandable carrier material and adhering the semiconductor wafer to a surface of the carrier material. The method may also include applying a tensile force on the wafer by expansion of the carrier material to form cracks in the streets between the microelectronic devices and separate the microelectronic devices.

(40) In some embodiments, as illustrated in FIG. 8, the wafer **202** may be inverted on the carrier wafer **304** and transferred to a carrier material **802**, such as dicing tape or mount tape or film. To transfer the wafer **202** from the carrier wafer **304** to the carrier material **802**, the adhesive **306** may be released to release the wafer **202** from the carrier wafer **304**. For example, the adhesive **306** may be released through a light or laser impinging on the wafer **202** or carrier wafer **304**. In other embodiments, the adhesive **306** may be released through other known methods, such as chemical releases and/or mechanical releases. The carrier material **802** may be secured to and supported by a film frame **804**. The carrier material **802** may include an adhesive **806** formulated to secure the wafer **202** to the carrier material **802**. The rear surface **302** of the wafer **202** may be adhered to the carrier material **802**, such that the active surface **208** of the wafer **202** faces upwardly and away from the carrier material **802**.

(41) The carrier material **802** may be a flexible material, expandable in the X-Y plane (i.e., laterally). After the wafer **202** is secured to the carrier material **802**, the carrier material **802** may be stretched by the film frame **804**. Stretching the carrier material **802** may apply tensile forces on the wafer **202**. The stress concentrations around the damage in the implanted regions **602** may cause the wafer **202** to fracture and cleave along the implanted regions **602** of streets **206**, creating cracks **702** through the wafer **202** that substantially coincide with the implanted regions **602** and/or the streets **206**. The cracks **702** may substantially separate the wafer **202** into individual semiconductor dice **204**.

(42) Larger amounts of damage in the implanted regions **602** may enable the cracks **702** to be formed with less tensile force. Thus, implanting the implanted regions **602** with ions from larger species may enable the cracks **702** to form under less tensile force. Reducing the tensile force may enable lighter materials to be used for the carrier material **802** and may reduce the amount thickness and strength of the adhesive required on the carrier material **802**. More options of carrier material **802** and adhesives may reduce the manufacturing costs, as carrier material **802** will not have to support stresses incurred by conventional blade dicing or heat from laser or stealth dicing. Further, carrier material **802** may provide advantages in chemical resistance, use at lower temperatures and thinner adhesives to provide better fixed positioning of the semiconductor dice **204**. Suitable carrier materials include KAPTON® polyimide film from DuPont Corporation, as well as various adhesive films available from NITTO Americas and LINTEC Corporation.

(43) In some embodiments, the wafer **202** may be separated through both a heating process as illustrated in FIG. 7 and the stretching process illustrated in FIG. 8. For example, the wafer **202** may be heated on the carrier wafer **304** as illustrated in FIG. 7, causing fractures and cracks **702** to propagate through the wafer **202**. After the cracks **702** are formed in the heating process, the wafer **202** may be transferred to the carrier material **802** and stretched. Stretching the wafer **202** may complete any incomplete fractures or cracks **702**, such as in areas where the concentration of damage and ions was insufficient to fracture completely through the thickness of the wafer **202** at the temperature used in the heating process. In some embodiments, a significantly lower temperature may be used on carrier wafer **304** to reduce a risk of temperature damage to the wafer **202**, such that the heating process may expand the damage and form some cracks **702**, which the

stretching process of FIG. 8 may complete any incomplete cracks 702.

(44) In some embodiments, the wafer 202 may be heated after being transferred to the expandable carrier material 802. For example, some types of carrier materials 802 may be selected to withstand relatively high temperatures, such that the wafer 202 may be heated while secured to the expandable carrier material 802. In some cases, the wafer 202 may then be stretched on the expandable carrier material 802 to complete any unfinished cracks 702 before picking the dice 204 from the surface of the carrier material 802. In other cases, the dice 204 may be picked from the surface of the carrier material 802 without further stretching the wafer 202.

(45) In some embodiments, the wafer 202 may be completely singulated through only one of the above processes. For example, the cracks 702 may be formed through the heating process of FIG. 7 and upon completing the heating process the individual dice 204 may then be picked from the surface of the carrier wafer 304 or expandable carrier material 802 for further processing, such as stacking. In some embodiments, the wafer 202 may be transferred to the carrier material 802 without having undergone any heating and the wafer 202 may then be stretched as described above to singulate the wafer 202 into individual dice 204, which may then be picked from the carrier material 802 to undergo further processing.

(46) In implementing methods of the embodiments, successful cleaving of wafer 202 on carrier wafer 304 due to heating may be confirmed by optical inspection from above. If wafer 202 is cleaved on expandable carrier material 802, carrier material may be illuminated from below and any uncleaved areas detected optically as opaque or partially occluded.

(47) Some embodiments of the present disclosure may include a microelectronic device 900 (e.g., semiconductor die). The microelectronic device 900 may include an active surface and a rear surface opposite the active surface. The microelectronic device may further include side surfaces of the semiconductor material extending between the active surface and the rear surface. The side surface may include a first portion having highly irregular (i.e., jagged) surface topography. The highly irregular surface topography may extend to a distance of between about 2 μm and about 6 μm from at least one of the active surface 208 and the rear surface 302 of the microelectronic device, depending upon the embodiment employed to singulate the microelectronic device 900 from a wafer 202. The side surface may further include a second portion extending from the first portion to another of the active surface and the rear surface having a less uneven surface.

(48) FIG. 9 illustrates a plan view of a side surface 906 of a microelectronic device 900 singulated by the method described above. The side surface 906 of the microelectronic device 900 may exhibit damage 908 from the implanted ions. As described above, the damage 908 may include dislocations and point defects resulting from the impact of the individual ions in the implant process. The damage 908 may show as peaks and valleys in the side of the microelectronic device 900 extending from the ion-implanted surface of the die 900 at least to an area of maximum species penetration. The side surface 906 may also include fracture lines 910 extending from the jagged or broken edges of the damage 908. The fracture lines 910 may be the result of the cracks 702 described above that are formed by heating and/or stretching the wafer 202.

(49) The side surface 906 of the microelectronic device 900 may also include residual ions 912 embedded within the side surface 906, such that the residual ions 912 may be detected, such as through secondary ion mass spectrometry (SIMS) or energy-dispersive X-ray spectroscopy (EDX) from the side surface 906 and may not be on the outer portion of the side surface 906 connected to the fracture lines 910. Some of the portions of damage 908 and/or residual ions 912 may be larger than others. For example, as described above, some of the ion species may be larger, such as boron, phosphorus, arsenic and some may be smaller, such as hydrogen. In some embodiments, the implant process may use multiple different ion species to achieve damage at different depths. The larger damage 908 and/or residual ions 912 may be closer to a first surface 902 than the smaller damage 908 and/or residual ions 912. The first surface 902 may be the surface into which the ions were implanted that is adjacent and transverse to the side surface 906.

(50) In some embodiments, only one species may be used. Some of ions may travel a greater distance into the wafer while others may experience more early collisions and may stop closer to the first surface **902**. The ions that travel farther into the wafer may experience collisions as they pass through the areas of the wafer closer to the first surface **902**. Therefore, the larger amounts of damage **908** and/or number of residual ions **912** may remain closer to the first surface **902**, with the amount of damage and/or number of residual ion **912** gradually reducing as the distance from the first surface **902** increases. Such an approach may result in a wedge effect, enhancing the potential for cleaving at lower temperatures, lower tensile stresses, or both.

(51) The damage **908** and/or residual ions **912** may extend to a distance from the first surface **902** of between about 1 μm and about 6 μm , such as between about 2 μm and about 5 μm , or between about 2 μm and about 4 μm .

(52) As described above, the ions may be implanted into the active surface **208** or the rear surface **302** of the wafer **202**. Therefore, in some embodiments, the first surface **902** may coincide with the active surface **208** of the wafer **202** and associated semiconductor dice **900**. In other embodiments, the first surface **902** may coincide with the rear surface **302** of the wafer **202** and associated semiconductor dice **900**. A second surface **904** on an opposite side of the microelectronic device **900** may be the opposing active surface **208** or rear surface **302** from the first surface **902** of the wafer **202** and associated semiconductor die **900**. The distance between the first surface **902** and the second surface **904** may be substantially the same as the final thickness of the associated wafer **202** after the thinning process illustrated in FIG. 4. Therefore, the distance between the first surface **902** and the second surface **904** may be less than about 30 μm , such as less than about 20 μm , or less than about 10 μm .

(53) FIG. 10 illustrates a profile view of a side surface **906** of the semiconductor die **900**. The portion of the side surface **906** where the damage **908** is present may exhibit a highly irregular topography **1002** characterized by sharp edges, protrusions, and recesses surrounding the points of damage **908** where the ions collided with the materials of the wafer as they were implanted into the first surface **902** of the wafer **202**. Thus, the highly irregular topography **1002** may begin proximate the first surface **902** of the microelectronic device **900** extending toward the second surface **904**. The highly irregular topography **1002** may extend a distance from the first surface **902** of between about 1 μm and about 6 μm , such as between about 2 μm and about 5 μm , or between about 2 μm and about 4 μm .

(54) The remainder of the side surface **906** may be a somewhat irregular, but less uneven surface **1004**, characterized by flat surfaces interrupted by fracture lines **910**. The uneven surface **1004** may extend from the highly irregular topography **1002** to the second surface **904**.

(55) As described above, residual ions **912** may also be imbedded in the microelectronic device **900** to a depth beneath the side surface **906** of the microelectronic device **900**. The residual ion **912** may be located in the microelectronic device **900** in substantially the same regions as the highly irregular topography **1002** and laterally adjacent the highly irregular topography **1002**.

(56) FIG. 11 illustrates a top view of the microelectronic device **900**. The microelectronic device **900** may include an active region **1106** where integrated circuitry and associated features of the microelectronic device **900** may be formed. The active region **1106** of the microelectronic device **900** may be substantially surrounded by a barrier **1104** extending downwardly from the surface including active region **1106** into the semiconductor material of the microelectronic device **900** and configured to separate the active region **1106** from side surfaces **906**, which may form the outer perimeter of the microelectronic device **900**. The barrier **1104** may be formed from a material selected to substantially prevent diffusion of other materials, such as the ions from the implant process, into peripheral areas of the active region **1106** bearing integrated circuitry. The barrier **1104** may be formed from a material such as tungsten, titanium, cobalt, ruthenium, tantalum, tantalum nitride, indium oxide, tungsten nitride, titanium nitride, etc. In some embodiments, the barrier **1104** may also be configured to mechanically protect the active region **1106**, such that when

the wafer is singulated any resulting cracks in the side surfaces **906** do not propagate into the active region **1106**.

(57) The side surfaces **906** may include residual ions **912** embedded from the implant process. The active region **1106** may be substantially free of residual ions **912**. As described above, the implant process may include a mask **502** configured to substantially control the locations where the implanted ions are directed to coincide with the streets **206**. Furthermore, as noted above, the barrier **1104** may be configured to substantially block any errant ions from entering the active region **1106** from semiconductor material adjacent to side surfaces **906**.

(58) Some embodiments of the present disclosure may include a microelectronic package. The microelectronic package may include one or more microelectronic devices. The microelectronic devices may each include side surfaces of semiconductor material. The side surfaces may include a first portion having a highly irregular topography. The highly irregular topography may extend to a distance of between about 2 μm and about 6 μm from an adjacent major surface of the microelectronic device. The side surface may further include a second portion adjacent the first portion and having a less uneven surface.

(59) FIG. 12 illustrates a microelectronic device **1200** formed from a stack of semiconductor dice **1202**. The sides of the stack of semiconductor dice **1202** may be characterized by a pattern of alternating highly irregular surface topographies **1208** and uneven surfaces **1204**. As shown, semiconductor dice **1202** were fabricated using implantation of a wafer **202** from a rear surface **302** thereof, resulting in highly irregular surface topographies **1208** on the upper portions of sides **1206** of semiconductor dice **1202**.

(60) The semiconductor dice **1202** may be stacked on a substrate **1212** configured to electrically couple the stack of dice **1202** to another component. The substrate **1212** may include discrete connection elements in the form of solder bumps **1210** configured to connect to higher level packaging. The substrate **1212** may be electrically coupled to the stack of die **1202**, such as through TSVs (not shown) formed through each semiconductor die **1202**. Similarly, each semiconductor die **1202** may be electrically coupled to the adjacent dice **1202** through the TSVs and conductive elements extending through the bond lines between adjacent semiconductor dice **1202** of the stack. For example, conductive (e.g., copper) pillars may connect to aligned terminal pads of adjacent semiconductor dice through solder reflow or diffusion bonding to operably couple the TSVs of adjacent dice **1202**.

(61) Embodiments of the present disclosure may enable the streets between die locations on a wafer to be reduced in width by requiring a much smaller area to complete the dicing operations to singulate a wafer into individual dice. Reducing the street width between the die locations on the wafer may enable a larger number of dice to be fabricated from a single wafer of semiconductor material.

(62) Furthermore, embodiments of the present disclosure may substantially decrease the organic (e.g., polymer film residue, adhesive residue) and inorganic (e.g., semiconductor material) contaminant particles produced during the process of dicing or singulating individual semiconductor dice from a wafer. Residual contaminant particles on individual semiconductor dice may, as previously noted, cause damage and/or failures in associated microelectronic devices. Thus, reducing the volume of contaminant particles produced during the dicing or singulation process may similarly decrease the potential for damage or failures of the dice and associated microelectronic devices. Reducing the number of damaged or failed dice or microelectronic devices and decreasing the space required between the individual dice on each wafer may increase the yield and reliability of semiconductor dice and microelectronic devices incorporating such dice.

(63) Increasing the yield of microelectronic devices by reducing the number of compromised semiconductor dice or assemblies and packages of same may increase the yield of such assemblies and packages, resulting in greater profitability from reduced costs for the production of the associated microelectronic devices. The microelectronic devices may be included in multiple

different types of electronic devices, such as personal electronics (e.g., mobile devices, phones, tablets, etc.), computers (e.g., personal computers, laptops, etc.), etc. Reducing the cost of producing the microelectronic devices may in turn reduce the cost of producing the associated electronic devices.

(64) The embodiments of the disclosure described above and illustrated in the accompanying drawing figures do not limit the scope of the invention, since these embodiments are merely examples of embodiments of the invention, which is defined by the appended claims and their legal equivalents. Any equivalent embodiments are intended to be within the scope of this disclosure. Indeed, various modifications of the present disclosure, in addition to those shown and described herein, such as alternative useful combinations of the elements described, may become apparent to those skilled in the art from the description. Such modifications and embodiments are also intended to fall within the scope of the appended claims and their legal equivalents.

Claims

1. A microelectronic device, comprising: a semiconductor material comprising: a perimeter region including residual ions and one or more of point defects and dislocations resulting from ion implantation extending from an outer edge of the perimeter region; and an active region substantially free from point defects and dislocations caused by ion implantation; and a barrier positioned between the active region and the perimeter region of the semiconductor material.
2. The microelectronic device of claim 1, wherein the barrier is formed from a material configured to substantially prevent diffusion of ions from the ion implantation.
3. The microelectronic device of claim 2, wherein the material of the barrier comprises one or more of tungsten, titanium, cobalt, ruthenium, tantalum, tantalum nitride, indium oxide, tungsten nitride, and titanium nitride.
4. The microelectronic device of claim 1, wherein the barrier is configured to protect the active region from crack propagation.
5. The microelectronic device of claim 1, wherein the barrier extends from a surface including the active region into the semiconductor material.
6. The microelectronic device of claim 1, wherein the active region includes integrated circuitry.
7. The microelectronic device of claim 1, wherein a distance between the barrier and a perimeter edge of the perimeter region is less than about 10 micrometers (μm).
8. A microelectronic device, comprising: a semiconductor material including: an active surface comprising an active region and a perimeter region separated by a barrier, wherein the barrier extends from the active surface into the semiconductor material; a rear surface opposite the active surface; and side surfaces extending between the active surface and the rear surface, the side surfaces respectively comprising: a first portion having a relatively more irregular surface topography including residual ions, the relatively more irregular surface topography extending from one of the active surface and the rear surface; and a second portion extending from the first portion to another of the active surface and the rear surface having a relatively less irregular surface topography.
9. The microelectronic device of claim 8, wherein the first portion of the side surfaces comprises one or more of point defects and dislocations resulting from ion implantation.
10. The microelectronic device of claim 8, wherein the barrier comprises one or more of tungsten, titanium, cobalt, ruthenium, tantalum, tantalum nitride, indium oxide, tungsten nitride, and titanium nitride.
11. The microelectronic device of claim 8, wherein the active region comprises integrated circuitry.
12. The microelectronic device of claim 8, wherein the barrier extends around a perimeter of the active region.
13. The microelectronic device of claim 8, wherein a distance between the active surface and the

rear surface is less than about 30 micrometers (μm).

14. A method of manufacturing a microelectronic device, the method comprising: forming microelectronic devices on an active surface of a wafer; securing the wafer to a carrier wafer; forming a barrier extending into the wafer through the active surface of the wafer between active regions of the microelectronic devices and streets between the microelectronic devices; implanting ions to initiate dislocations in semiconductor material of the wafer along the streets between the microelectronic devices, the dislocations extending to a first distance from one of the active surface and a rear surface of the wafer, such that the semiconductor material is substantially free of dislocations beyond the first distance from one of the active surface and the rear surface of the wafer; thinning the wafer; and heating the wafer to form cracks along the streets from stress concentrations proximate the dislocations.

15. The method of claim 14, wherein implanting ions comprises implanting the ions through a mask having openings substantially aligned with the streets between the microelectronic devices.

16. The method of claim 15, wherein the mask comprises a same mask as used to fabricate features of the microelectronic devices.

17. The method of claim 14, wherein implanting ions includes implanting ions of one or more of hydrogen, helium, boron, phosphorus and arsenic.
